

Semantic Image Retrieval System – Evaluation Results

Dataset Information

- **Dataset:** Flickr30k Subset
 - **Evaluation Queries:** 2000
 - **Evaluation Script Output:** query_results.csv
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Evaluation Metrics

Precision@K

Precision@K measures how many of the **top K retrieved images are correct** for a given text query.

Metric	Value	Interpretation
Precision@1	0.581	Correct image is ranked first in 58.1% of queries
Precision@5	0.164	On average ~0.82 correct images in top 5 results
Precision@10	0.088	On average ~0.88 correct images in top 10 results

Interpretation

- The system retrieves the **correct image as the top result more than half the time**.
 - This indicates good alignment between **text and image embeddings**.
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Recall@K

Recall@K measures whether the **correct image appears within the top K results**.

Metric	Value	Interpretation
Recall@1	0.581	Correct image appears at rank 1 in 58.1% of queries
Recall@5	0.8215	Correct image appears in top 5 results in 82.15% of queries
Recall@10	0.884	Correct image appears in top 10 results in 88.4% of queries

Interpretation

- The system successfully retrieves the correct image **within the top 5 results for most queries**.
 - High recall indicates the system **rarely misses relevant images**.
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Mean Reciprocal Rank (MRR)

MRR: 0.6823

MRR evaluates **how early the correct result appears in the ranking**.

Example:

Rank	Reciprocal Score
1	1.0
2	0.5
3	0.33

A higher MRR means relevant results appear **closer to the top of the ranking**.

Interpretation

- An MRR of **0.68** indicates that correct images typically appear **between rank 1 and rank 2**.

Mean Average Precision (mAP)

mAP: 0.6823

Mean Average Precision measures **overall ranking quality across all queries**.

Typical interpretation:

mAP	Performance
< 0.3	Weak
0.4 – 0.5	Moderate
0.6+	Good
0.7+	Strong

Interpretation

- The system achieves **good ranking performance** with mAP $\approx$ **0.68**.
- This indicates strong semantic matching between text queries and images.

Overall System Performance

The evaluation results demonstrate that the **Semantic Image Retrieval System performs effectively**.

Key observations:

- **58% Top-1 Accuracy** – The correct image is retrieved as the first result in more than half of the queries.
- **82% Recall@5** – The correct image appears within the top 5 results for most queries.
- **88% Recall@10** – The correct image is almost always retrieved within the top 10 results.
- **mAP = 0.68** – Indicates good ranking quality for retrieved results.

Overall, the system shows **strong semantic understanding between text queries and image content**, making it suitable for practical image retrieval applications.

Output Files

Detailed per-query evaluation results are stored in: